Homework 10

November 6, 2026

Problem 1

Prove the following theorem (Theorem 4.24 in the textbook).

Theorem 4.24. Self adjoint, skew self adjoint and unitary transformations are all normal.

Problem 2

Prove the following theorem (Theorem 4.25 in the textbook).

Theorem 4.25. Let $A \in \mathcal{C}^{n\times n}$. Then A is a normal matrix if and only if there exists a unitary matrix $U \in \mathcal{U}^{n\times n}$ such that

$$U^*AU = \operatorname{diag}\{\lambda_1, \lambda_2, \ldots, \lambda_n\},$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A.

Problem 3

Prove the following theorem (Theorem 4.27 in the textbook).

Theorem 4.27. Let $A \in \mathcal{C}^{n\times n}$ with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then

$$\sum_{i=1}^{n} |\lambda_i|^2 \leqslant \sum_{i=1}^{n}\sum_{j=1}^{n} |a_{ij}|^2,$$

and the equality holds if and only if A is a normal matrix.

Problem 4

For the given matrix

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix},$$

find the unitary matrix W such that W^*AW is upper triangular.

Problem 5

Prove the following theorem (Theorem 4.28 in the textbook), following the same line of reasoning as the proof of Theorem 4.23.

Theorem 4.28. Let $A \in \mathcal{C}^{n\times n}$. A is a normal matrix if and only if $AA^* = A^*A$.

Requirement. Start only from the definition of a normal matrix, and do not quote any other theorem or fact.

Problem 6

Prove the following theorem (Theorem 4.29 in the textbook).

Theorem 4.29. Let $A \in \mathcal{C}_n^{n\times n}$. Then A can be uniquely factorized as

$$A = UR,$$

where $U \in \mathcal{U}^{n\times n}$, R is an upper triangular matrix with positive diagonal elements.

Problem 7

Prove the following theorem (Theorem 4.30 in the textbook).

Theorem 4.30. Let $A \in \mathcal{C}^{n\times n}$ be a normal matrix, then

(1) A is Hermitian if and only if all eigenvalues of A are real.

Problem 8

Prove the following theorem (Theorem 4.31 in the textbook).

Theorem 4.31. Let $A \in \mathcal{C}^{n\times n}$, then

(1) A is Hermitian if and only if for any $x \in \mathcal{C}^n$, x^*Ax is real.

Problem 9

Prove the following property. (It is property 5 in the list of properties of Hermitian matrices, page 168 of the textbook.)

Property 5. For Hermitian matrices A and B, AB is Hermitian if and only if $AB = BA$.